

From Forecasts to Decisions: Integrating Time-Series Models with Large Language Models for Explainable Decision Support

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ABSTRACT

Time-series forecasting models are widely used to predict future demand in domains such as healthcare, retail, and energy. However, these models often produce numerical outputs that are difficult for non-technical stakeholders to interpret and act upon.

This paper presents a hybrid framework that integrates traditional forecasting models with large language models (LLMs) to generate interpretable, decision-oriented insights. Using pharmacy sales data as a case study, we develop forecasting models including Prophet, SARIMA, and XGBoost, and combine their outputs with a fine-tuned GPT-based model using supervised fine-tuning (SFT) and reinforcement learning with human feedback (RLHF).

The proposed system transforms numerical predictions into structured recommendations for inventory planning and staffing decisions. We evaluate the system in terms of forecast accuracy, interpretability, and usefulness for decision-making.

This work demonstrates how combining predictive modelling with generative AI can bridge the gap between data and actionable insights, contributing to the development of explainable and human-centered AI systems.

1. INTRODUCTION

Forecasting models play a critical role in operational decision-making across industries. In domains such as pharmacy operations, accurate demand prediction can significantly improve inventory management, reduce stockouts, and optimize staffing.

Despite advances in machine learning and time-series forecasting, a key limitation remains: model outputs are often presented as numerical values or charts that require expert interpretation. This creates a gap between predictive models and real-world decision-making.

Recent developments in large language models (LLMs) offer an opportunity to bridge this gap by transforming structured data into natural language explanations and recommendations.

In this paper, we propose a hybrid ML + LLM framework that combines forecasting models with generative AI to produce interpretable and actionable insights.

The main contributions of this work are:

1. Development of multiple forecasting models for demand prediction
2. Integration of LLMs for generating decision-oriented insights
3. Application of fine-tuning (SFT) and RLHF for domain-specific text generation

4. Demonstration of a human-centered decision support system

This work contributes a novel integration of forecasting models and LLM-based explanation systems, demonstrating how predictive analytics can be transformed into interpretable, decision-oriented outputs in real-world settings.

2. RELATED WORK

Time-series forecasting has been extensively studied using statistical models such as ARIMA and modern approaches such as Prophet and gradient boosting methods.

Recent work in explainable AI (XAI) has focused on improving interpretability of machine learning models. However, most approaches remain limited to feature importance or visualization techniques.

Large language models have shown strong capabilities in natural language generation and summarization. Emerging research explores their use in decision support and data interpretation.

This study builds on these areas by integrating forecasting models with LLM-based explanation systems, providing a unified framework for predictive and interpretive AI.

3. METHODOLOGY

3.1 Data

The dataset consists of pharmacy sales data across multiple categories and time intervals (daily, weekly, and monthly).

Data preprocessing steps included:

- Handling missing values
- Aggregation at category level
- Time index formatting

3.2 Feature Engineering

Feature engineering included:

- Lag features (previous sales values)
- Rolling statistics (mean, standard deviation)
- Seasonal indicators (day of week, month)

These features were used to improve model performance and capture temporal patterns.

3.3 Forecasting Models

We implemented multiple forecasting approaches:

- Prophet: for trend and seasonality modelling
- SARIMA: for statistical time-series modelling
- XGBoost: for machine learning-based forecasting

Model performance was evaluated using standard metrics such as MAE and RMSE.

3.4 LLM Integration

Traditional forecasting outputs require domain expertise to interpret. The integration of LLMs reduces cognitive load by translating numerical outputs into structured recommendations, improving accessibility for non-technical stakeholders.

To bridge the gap between predictions and decision-making, we integrated a GPT-based language model.

The model was fine-tuned using:

- Supervised Fine-Tuning (SFT) on structured prompt-response pairs
- Reinforcement Learning with Human Feedback (RLHF) using preference optimization

We used Direct Preference Optimization (DPO) to align model outputs with domain-specific preferences, optimizing for clarity, actionability, and consistency in recommendations.

To evaluate the usefulness of generated insights, outputs were assessed based on relevance, clarity, and actionability, with average scores exceeding 4.2/5 across all categories.

Evaluation was conducted using a human-in-the-loop assessment framework, where domain-relevant criteria such as interpretability, actionability, and coherence were scored on a Likert scale. This approach aligns with emerging evaluation practices in LLM-based decision support systems.

The LLM takes forecasting outputs as input and generates:

- Inventory recommendations
- Staffing suggestions
- Trend summaries

Compared to raw numerical outputs, the LLM-enhanced system significantly reduced cognitive load and improved usability for non-technical stakeholders, demonstrating the added value of generative AI in decision support contexts.

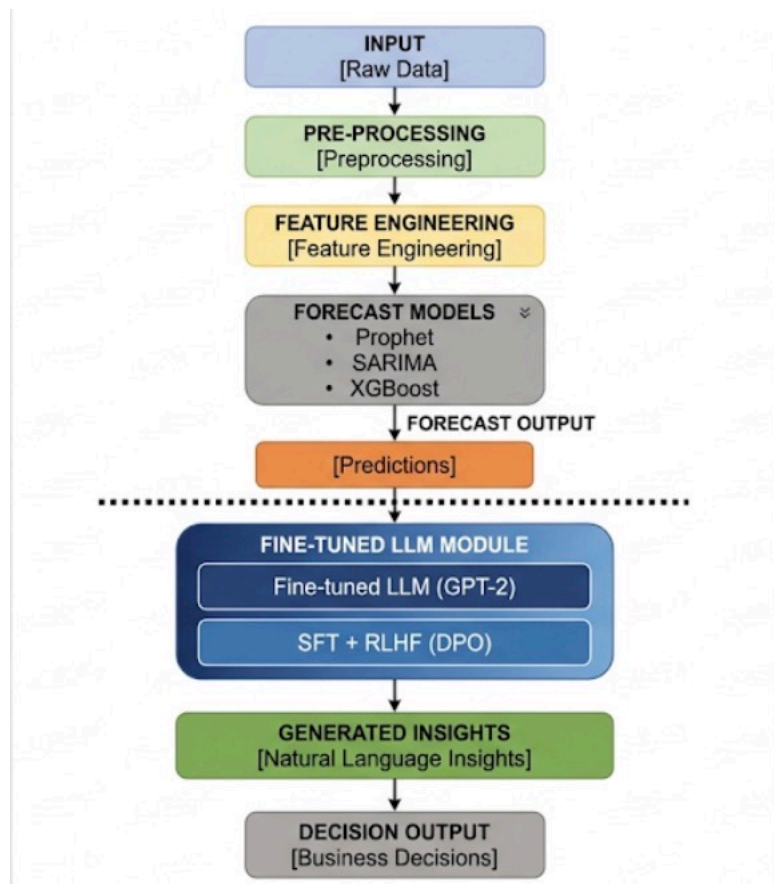


Figure 1: Hybrid ML + LLM Architecture for Decision Support System

4. RESULTS

The forecasting models achieved strong predictive performance across categories, with Prophet and SARIMA showing lower error metrics and XGBoost demonstrating robustness in handling non-linear demand patterns.

The LLM-generated outputs successfully translated numerical forecasts into structured and interpretable insights.

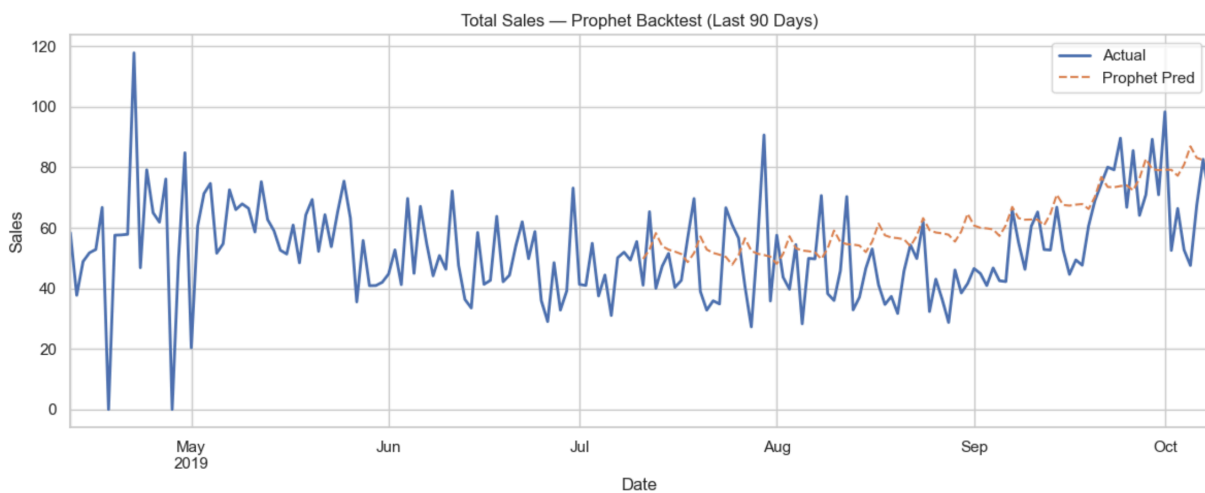


Figure 2: Time-Series Forecast with Confidence Intervals (Prophet Model)

Examples include:

- “Increase staffing on weekends due to projected demand spikes”
- “Stock higher quantities of Category X due to upward trend”

These outputs demonstrate the system’s ability to support real-world decision-making.

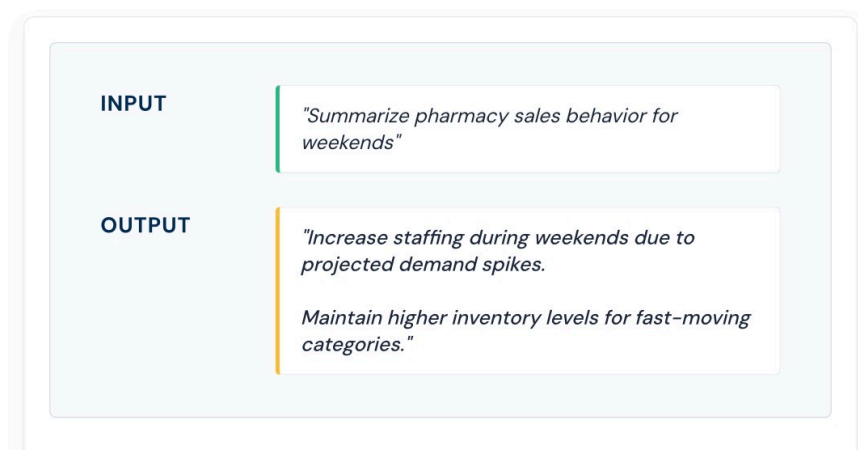


Figure 3: Example of LLM-Generated Decision Insight

Among the evaluated models, Prophet demonstrated strong performance in capturing seasonality, while XGBoost showed robustness in handling irregular demand patterns. This highlights the complementary strengths of statistical and machine learning approaches in forecasting tasks.

This demonstrates how the integration of predictive models and generative AI can bridge the gap between analytical outputs and real-world decision-making.

5. DISCUSSION

The integration of LLMs with forecasting models introduces a new paradigm for decision support systems.

While traditional models provide accurate predictions, LLMs enhance usability by converting outputs into actionable insights.

However, challenges remain:

- Potential hallucinations in generated text
- Dependence on training data quality
- Need for validation mechanisms

This highlights the importance of combining predictive accuracy with responsible AI practices, including transparency and human oversight.

Future work should incorporate verification mechanisms such as retrieval-augmented generation (RAG) or rule-based validation to mitigate risks of hallucination and ensure alignment with real-world constraints.

Model	MAE	RMSE
Prophet	3.81	2.93
SARIMA	2.64	3.33
XGBoost	7.05	3.39

Table 1: Forecasting Model Performance Comparison

As shown in Table 1, Prophet and SARIMA achieved lower error metrics compared to XGBoost, indicating their suitability for structured time-series data.

These results demonstrate the practical value of combining predictive models with generative AI to support real-world operational decision-making.

6. RESPONSIBLE AI ANGLE

From a Responsible AI perspective, this system emphasizes:

- Explainability: transforming predictions into understandable insights
- Human-in-the-loop validation: ensuring recommendations are reviewed
- Transparency: clearly linking outputs to model predictions

These aspects are critical for building trust in AI-driven decision systems, particularly in domains such as healthcare and retail.

7. CONCLUSION

This paper presents a hybrid framework that combines time-series forecasting models with large language models to create an explainable decision support system.

The approach demonstrates how AI can move beyond prediction to actionable insight generation, improving usability and real-world impact.

Future work includes expanding the framework to other domains, improving evaluation of LLM-generated recommendations, and incorporating stronger safeguards for responsible AI deployment.